Big Data Analytics

Irene Siragusa, PhD

CHILAB
Università degli Studi di Palermo

Outline

- General information
- Pre-requisites
- Course objectives
- Topics
- Textbooks
- Tools

General information

Big Data (12 ECTS)

Big Data Tools

- 1st semester
- 6 ECTS course

Big Data Analytics

- 2nd semester
- 6 ECTS course

General information

Big Data (12 ECTS)

Big Data Tools

- 1st semester
- 6 ECTS course

Big Data Analytics

- 2nd semester
- 6 ECTS course

✨ it depends ✨

✨ it's a tradeoff ✨

General information

Lectures, where and when?

- 08-10 Thursday @ room SEAS 2B building 13
- 10-12 Friday @ room SEAS 2B building 13

General information

?Contact me?

- irene.siragusa02@unipa.it w/ subject [BIG-DATA]

?Meet me?

- Tuesday from 15:00 to17:00 @ CHILab, building 6 in Viale delle Scienze, entrance 6A, 3rd floor, first door on the left
- Phone: +3909123862643
- But an e-mail is preferred

General information

- *Course material?*
  - BIG-DATA-ANALYTICS GitHub https://github.com/iresiragusa/BIG-DATA-ANALYTICS/
    - Slides -> theory slides
    - Exercises -> python notebooks

- Subscribe from the university website to the course for official communications

General information

Exams?

- BIG-DATA C.I.
  - Written and oral exam with topics of both Big Data Tools and Big Data Analytics
- BIG-DATA ANALYTICS (erasmus students)
  - Written and oral exam with Big Data Analytics topics
- BIG-DATA ANALYTICS (PhD students)
  - Oral presentation on the usage of Big Data Analytics topics in their reaseach

Pre-requisites

- Basic knowledge of probability, statistics, and linear algebra
- Basic Python programming
- Big Data Tools
  - From https://github.com/iresiragusa/BIG-DATA-TOOLS
    - Slides > from 01 to 04
    - Execises > from notebook_practice 1 to notebook_practice 5b
    - Exams > Exams/COMPITO BDT demo.pdf exercise 1

Course objectives

- The course is aimed at providing a deep knowledge of the main algorithms for data analysis both in Big Data perspective involving classic Machine Learning and Deep Learning strategies.
- The objective it to be able to build autonomously a complete data analysis pipeline for real problems.
- This course is arranged in lessons and exercise sections.

Any doubts? Just ask (I'll do my best in answering)

Course objectives

Design a suitable Big Data Architecture

- Being aware of numerical and statistical characteristics of different data types
- Determine the appropriate acquisition and pre-processing phases
- Determine the appropriate analysis and prediction processes over data
  - ML vs DL
  - Spark ML Pipeline
  - Pytorch
- Leveraging statistical characteristics and Python

Course objectives - Theory

- Information theory and Probability
- Machine Learning
  - Supervised and Unspervised Learning
  - Reinforcement Learning
  - Clustering and Classification
- Deep Learning
  - Deep Learning foundamentals
  - CNN, Autoencoders, LSTM, GAN, GNN, Transformers

Course objectives - Practical

- Clustering w/ sci-kit learn
- Classification w/ sci-kit learn
- Spark ML pipeline for clustering
- Spark ML pipeline for classification
- Neural Network in PyTorch

Textbooks

- Data Mining: The Textbook, 2015, Charu C. Aggarwal, Springer-Verlag New York, ISBN 978-3319141411, https://link.springer.com/book/10.1007/978-3-319-14142-8
- Deep Learning, 2016, Ian Goodfellow, Yoshua Bengio, Aaron Courville, MIT Press, ISBN 978-0262035613

  https://www.deeplearningbook.org/
- Spark: The Definitive Guide: Big Data Processing Made Simple, 2018, Bill Chambers e Matei Zaharia, Oreilly & Associates Inc, ISBN 978-1491912218

Textbooks

- Python Machine Learning: Machine Learning and Deep Learning with Python, scikit-learn,
- PyTorch Computer Vision Cookbook, https://github.com/akansh12/Computer_vision_Boxok_solution/
- Lecture notes

Topics

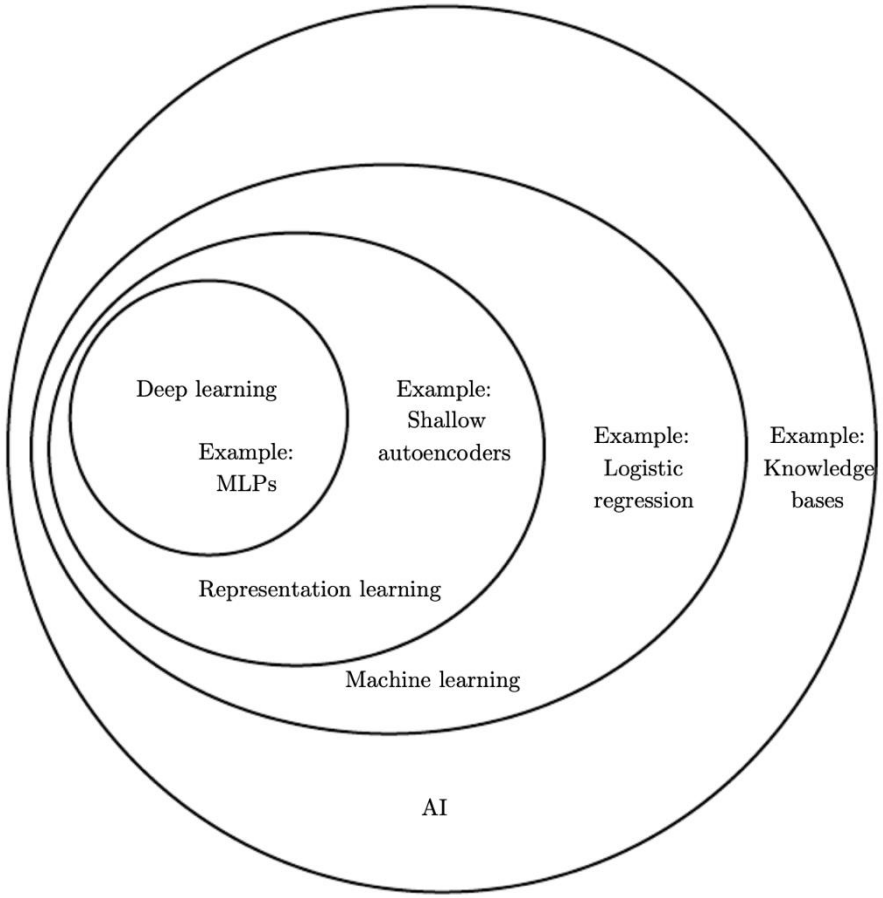
- Probability and Information Theory
- Sampling strategies
- Machine Learning
  - Supervised and Unsupervised Learning, Reinforcement Learning, Model Capacity, Parameters and Hyperparameters, Error Types, Training.
- Clustering
  - k-means algorithm and similar approaches, hierarchical clustering, density-based and grid-based clustering, high dimensional data, clustering evaluation, outliers.

Topics

- Classification basics
  - Feature selection, decision trees, rule-based classifiers, Naïve Bayes, Logistic Regression, Support Vector Machines, Nearest Neighbor, classifiers, evaluation.
- Classification advanced
  - Multi-class and rare class learning, regression on numeric data, Semi-supervised learning, ensemble methods.
- Deep learning
  - Neural Network, hidden and output units, loss functions, computational graph, stochastic gradient descent, optimization and regularization, CNN, Autoencoders, LSTM, GAN, Graph Neural Networks, Transformers, fine tuning and transfer learning.

AI? Machine Learning? Deep Learning?

AI? Machine Learning? Deep Learning?



Tools

- VSCode
  - https://code.visualstudio.com/download
- Python ≥ 3.7
  - [W10-W11] Microsoft store
  - [Linux] packet manager (apt, snap, …)
  - [MacOS] brew
  - [all] https://www.python.org/downloads/
- Google Colab
  - https://colab.google/